1.51 Bivariate Data: Association and Spearman's Rank

Worked Solutions

1)

Positive nonlinear association

2)

Positive nonlinear association

3 a)

$$\sum d_i^2 = 4 + 0 + 1 + 0 + 1 + 4 + 1 + 1 = 12$$

$$n = 8$$

$$r_s = 1 - \frac{6 \times 12}{8(8^2 - 1)} = \frac{6}{7} = 0.857 \text{ (3 SF)}$$

3 b)

Strong Positive

4)

x	4.3	3.8	8.6	8.0	1.1	5.5	4.8	6.5
y	7.9	8.1	7.3	7.2	9.3	7.7	7.8	7.6
Rank x	3	2	8	7	1	5	4	6
Rank y	6	7	2	1	8	4	5	3
d	-3	-5	6	6	-7	1	-1	3
d^2	9	25	36	36	49	1	1	9

$$\sum d_i^2 = 9 + 25 + 36 + 36 + 49 + 1 + 1 + 9 = 166$$

$$n = 8$$

$$r_s = 1 - \frac{6 \times 166}{8(8^2 - 1)} = -\frac{41}{42} = -0.976 \text{ (3 SF)}$$

5)

Student	A	B	C	D	E	F	G	H
x	2	3	4	5	1	6	3	2
y	49	58	62	60	59	71	46	66
Rank x	2.5	4.5	6	7	1	8	4.5	2.5
Rank y	2	3	6	5	4	8	1	7
d	0.5	1.5	0	2	-3	0	3.5	-4.5
d^2	$\frac{1}{4}$	$\frac{9}{4}$	0	4	9	0	$\frac{49}{4}$	$\frac{81}{4}$

$$\sum d_i^2 = \frac{1}{4} + \frac{9}{4} + 0 + 4 + 9 + 0 + \frac{49}{4} + \frac{81}{4} = 48$$

$$n = 8$$

$$r_s = 1 - \frac{6 \times 48}{8(8^2 - 1)} = \frac{3}{7} = 0.429 \text{ (3 SF)}$$